



Cambridge IGCSE™

MATHEMATICS

0580/23

Paper 2 Extended

October/November 2022

MARK SCHEME

Maximum Mark: 70

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2022 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **6** printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Maths-Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

Abbreviations

cao – correct answer only

dep – dependent

FT – follow through after error

isw – ignore subsequent working

oe – or equivalent

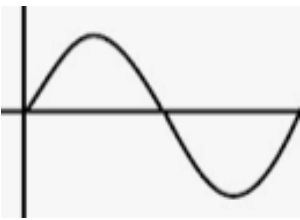
SC – Special Case

nfw – not from wrong working

soi – seen or implied

Question	Answer	Marks	Partial Marks
1	5[h] 23[min]	1	
2(a)	121	1	
2(b)	216	1	
3	6.05 or 6.054 to 6.055	1	
4	93 95 101 101	3	M1 for 4×97.5 implied by 390 or for four numbers which add to 390 B1 for four numbers with a range of 8 B1 for four numbers with mode of 101 to a maximum of 2 marks
5	$\frac{15}{21}$ and $\frac{14}{21}$ oe	M1	Allow any correct common denominator 21k
	$\frac{1}{21}$ cao	A1	
6(a)	$\frac{7}{20}$ oe or 0.35 or 35%	2	M1 for $1 - \left(\frac{2}{5} + \frac{1}{4}\right)$ oe
6(b)	48	1	
7	180	2	M1 for answer $2 \times 2 \times 3 \times 3 \times 5$ or better or for answer 180k or two correct factor trees, tables or Venn diagram or better or a list of multiples of both 36 and 60 with at least 3 correct of each
8	(1, 3.5)	2	B1 for each
9	[x =] 9 [y =] 3	2	B1 for each answer
10(a)	9.8[0] or 9.797 to 9.798	3	M2 for $14^2 - 10^2$ oe or better or M1 for $10^2 + h^2 = 14^2$ oe or better
10(b)	33.8 or 33.79 to 33.80	1	FT 24 + <i>their</i> (a)
11	15	4	B2 for $x = 16$ soi or M1 for $7x + 44 + x + 8 = 180$ or better M1 for $360 \div (\text{their } x + 8)$ oe
12	320.18	3	B2 for 4320.18 or M1 for $4000 \times \left(1 + \frac{2.6}{100}\right)^3 [-4000]$ oe

Question	Answer	Marks	Partial Marks
13	$2.\dot{4} - 0.2\dot{4}$ oe	M1	
	$\frac{11}{45}$ cao	B1	If 0 scored SC1 for $\frac{k}{90}$.
14	49.6	2	M1 for answer figs 496
15(a)	2	1	
15(b)	25.125	4	M3 for $\frac{15 \times 30}{2} + 30(k - 15) [= \text{figs } 45]$ oe OR B2 for 44 775 or 44.775 OR M1 for $\frac{15 \times 30}{2}$ or $30(k - 15)$ oe B1 for 45 000 or 0.225 or 0.03
16	$[y =] -\frac{1}{4}x - \frac{11}{2}$ oe	3	M1 for $\text{grad} = -\frac{1}{4}$ oe soi M1 for correct substitution shown of $(-2, -5)$ into $y = (\text{their } m)x + c$ oe ($\text{their } m \neq 4$)
17	8	3	
18	16	3	M2 for $12 \times \sqrt[3]{\frac{768}{324}}$ oe or M1 for $\sqrt[3]{\frac{768}{324}}$ or $\sqrt[3]{\frac{324}{768}}$ or $\frac{h^3}{12^3} = \frac{768}{324}$ oe
19(a)	$\frac{2}{x-1}$ final answer	2	M1 for $\frac{10}{5x-3-2}$ or better
19(b)	$\frac{10}{x} + 2$ or $\frac{10+2x}{x}$ final answer	3	M2 for $y - 2 = \frac{10}{x}$ or $x = \frac{10+2y}{y}$ oe or $yx = 10 + 2x$ oe or M1 for $x = \frac{10}{y-2}$ or $y(x-2) = 10$ oe or better
19(c)	$x - 1$	1	

Question	Answer	Marks	Partial Marks
20(a)	 Correct sketch to go through (0, 0), (180, 0) and (360, 0)	2	B1 for correct sine curve shape through the origin
20(b)	187.2 and 352.8	3	B2 for one correct value, if more than two answers given award B2 if any of the correct answers found and may be in the working or M1 for $\sin x = -\frac{1}{8}$ oe soi If 0 scored, SC1 for two reflex angles with a sum of 540 or two non-reflex angles with a sum of 180
21	076 or 076.4 to 076.5	5	B3 for [angle ABC =] 144 or 144.4 to 144.5 OR M2 for $[\sin ABC =] \frac{17.6 \sin 25}{12.8}$ oe or M1 for $\frac{17.6}{\sin B} = \frac{12.8}{\sin 25}$ oe M1 for 180 – their 35.5 AND M1 for their angle ABC – (180 – 112) oe
22(a)	$2x^3 + x^2 - 25x + 12$ final answer	3	B2 for correct unsimplified expanded expression or for simplified four-term expression of correct form with 3 terms correct or B1 for correct expansion of 2 brackets with at least 3 terms out of 4 correct
22(b)	$\frac{2}{x}$ final answer	4	M1 for $\left[\frac{4}{2x-3} \right] \times \frac{2x^2 + 11x - 21}{2x^2 + 14x}$ oe soi B1 for $(x+7)(2x-3)$ oe factorised B1 for $2x(x+7)$ oe factorised